Xuanming Liu

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Internship Experience

- **Sphere Lab** Hong Kong, China
  Internship, Advised by Jinxiu Liu and Weiyang Liu *July 2025 - Present*
  Designed a novel distillation method for video generation models, achieving over 1.53× speedup compared to CausVid. Designed a method to "strengthen" the probability flow by imposing constraints in multiple dimensions of time and space, replacing the traditional "expanding" model scaling with a structured shortcut, thereby achieving efficient generation.

Research Interests

Generative AI, Efficient Multimodal Models
Overcoming the "efficiency-quality" dilemma of generative AI. The existing powerful generative models perform exceptionally well in offline tests, but their high inference costs severely hinder their application in key scenarios such as real-time interaction, mobile computing, and long content generation. My research interest lies precisely in addressing this core contradiction: Can we, and how, make generative AI maintain "good" performance when it must be "fast" and "light"? I explore a series of cutting-edge approaches beyond simple model compression, including dynamic computing allocation, essential optimization of the generation process, and collaborative design of training-inference, aiming to systematically break through the efficiency bottlenecks of existing models and unlock their new possibilities in areas such as game simulation, real-time creation, and edge computing.

Education

- **Nanjing Normal University Affiliated Middle School** Nanjing, Jiangsu, China
  Full-time student *Sep 2021 - Jun 2024*

Representative Projects

- Streaming Autoregressive Video Generation via Diagonal Distillation

  Under Review Paper Link

  Accepted by ICLR 2026, (*Top 10.75% Submission* according to *https://papercopilot.com/statistics/iclr-statistics/iclr-2026-statistics/*)

  Xuanming Liu*, Jinxiu Liu*, Kangfu Mei, Yandong Wen, Ming-Hsuan Yang, Weiyang Liu

  (alphabetical order in paper)

  - Summary of the paper:*[A novel streaming autoregressive video generation method named "Diagonal Distillation" has been proposed to address the issues of high computational cost, long sequence error accumulation, and temporal inconsistency in existing diffusion models for real-time applications. The core innovation lies in the diagonal denoising strategy, which adopts an asymmetric generation approach: when generating video segments, more denoising steps are assigned to earlier segments, while fewer steps are assigned to subsequent segments. This method enables later segments to inherit the rich structural prior information obtained from the fully processed earlier segments, thereby significantly reducing the total denoising steps while maintaining the visual quality and motion coherence of the video. The method also designs a diagonal forced training mechanism, by explicitly simulating this diagonal denoising path during training and aligning the training and inference processes, effectively alleviating the long-term error propagation. Additionally, the paper introduces a flow distribution matching loss to maintain the motion amplitude and dynamic consistency of the generated video under strict step limits. Experiments show that this method can generate 5-second videos at a rate of up to 31 FPS, accelerating by 277.3 times compared to the non-distilled model, and surpassing existing advanced methods in terms of quality, achieving a better balance between generation efficiency and quality.]*

- SymbOmni: Evolving Agentic Omni Models via Symbolic Concept Learning

  Under Review Paper Link

  CVPR 2026 Under Review, Top 17% Submission

  Jinxiu Liu*, Jianru Li*, Kuang Tanqing*,**Xuanning Liu**, Kangfu Mei, Yandong Wen, Weiyang Liu

  (alphabetical order in paper)

  - Summary of the paper:*[A framework named SymbOmni has been proposed, which aims to address the "eternal novice" problem of current visual generation models. Its core innovation lies in the introduction of the Symbolic Concept Box (CB), an optimizable symbolic memory module that can abstract low-level operations into reusable Symbolic Workflow Instructions (SWIs). This framework achieves continuous evolution through an "induction-transformation" cycle: in the*

induction stage, the system abstracts successful experiences into symbolic knowledge and stores them in the CB; in the transformation stage, it solves new tasks by retrieving and combining these symbolic concepts. Specifically, the system adopts a language-based backpropagation mechanism, converting execution feedback into language gradients to optimize the CB, thereby enabling self-evolution without gradient fine-tuning. Experiments show that SymbOmni outperforms existing multimodal models on benchmarks such as ComfyBench, achieving significantly better results while maintaining high-quality generation. Token consumption is reduced by more than 40% compared to existing models, and it demonstrates true cumulative learning capabilities.]

- XYZFlow: Scaling Multidimensional Shortcut Flows for Efficient Generative Modeling

  Under Review Paper Link

  ICML 2026 Under(5(Accept), 4(Weak Accept), 4(Weak Accept), About 5% Submission)

  Xuanning Liu[*], Jinxiu Liu[*], Kangfu Mei, Yandong Wen, Weiyang Liu

  (alphabetical order in paper)

  - Summary of the paper:*[A novel and highly efficient generative modeling framework named XYZFlow has been proposed. Unlike the mainstream acceleration methods that rely on improved distillation algorithms, the core idea of XYZFlow is to shift from "extensive" scaling to "intensive" scaling. This is achieved by introducing structured, multi-dimensional "conditional constraints" to enhance the expressiveness of the probability flow itself, making its trajectory more straight and unique. This framework implements scaling along two orthogonal dimensions: 1) The time dimension, where a complete denoising history is injected as a condition for each denoising step to establish a "time coordinate system"; 2) The spatial dimension, where a "next shortcut prediction" mechanism is proposed, dividing the image into grids and generating them sequentially. Subsequent image blocks not only rely on the final output of the previous block but also use its complete denoising trajectory as a strong prior, thereby constructing high-dimensional constraints in the spatial domain. XYZFlow adopts a progressive denoising scheduling approach. The first block uses a larger number of steps to establish an anchor point, and subsequent blocks gradually reduce the number of steps by leveraging enhanced context, achieving implicit enhancement of the probability flow and straightening of the trajectory.]*

Open source project

- Streaming Autoregressive Video Generation via Diagonal Distillation

  Sphere-AI-Lab/diagdistill

  Star:126

Collaborators

- **Jinxiu Liu** Ph.D. in Computer Science at Georgia Tech, Research Scientist at Nex-AGI Team, leading efficient foundation model and agentic system development. Previously research intern at ByteDance, Google DeepMind, and Stanford AI Lab.

- **Ming-Hsuan Yang** Computer scientist and professor at the University of California, Merced, and research scientist at Google DeepMind specializing in computer vision, machine learning, AI, and robotics. An IEEE, ACM, AAAI, and AAAS Fellow, he is a Highly Cited Researcher (2018–2025) known for influential work in object tracking, face detection, and image super-resolution. His honors include the NSF CAREER Award, Longuet-Higgins Prize, and ICML Best Paper Award.

- **Weiyang Liu** Assistant Professor, Department of Computer Science and Engineering, The Chinese University of Hong Kong (CUHK); Affiliated Researcher, Max Planck Institute for Intelligent Systems. Head of the SphereLab, focusing on principled machine learning, foundation models, and reasoning systems.

- **Kang-Fu Mei** Senior Research Scientist, Google DeepMind. Lead researcher on video generation (Veo/Genie) and diffusion models. Expertise includes multimodal generative AI, world modeling, and real-time on-device AI systems.

Skills

- Vibe coding
- LaTeX & Scientific Writing
- Visualization & Figure Design
- Fundamental Programming & Development(Python)